What the decomposition gives you

Once $A = QDQ^{\mathsf{T}}$ is in hand, every entry below is read off D alone. The orthogonal factor never changes — which is the whole reason functions of the matrix become functions of its eigenvalues.

7
READINGS

FUNCTIONS OF THE MATRIX — 4

1 **Powers**

$k \geq 0$ integer; any k if all $\lambda_i > 0$

$$A^k = QD^kQ^{\mathsf{T}}$$

The middle factors collapse: $QDQ^{\mathsf{T}}QDQ^{\mathsf{T}} = QD^2Q^{\mathsf{T}}$, since $Q^{\mathsf{T}}Q = I$. Raising a diagonal matrix to a power means raising each entry, so A^{100} costs no more than A^2 — the reason diagonalization is worth the effort at all.

2 **Inverse**

all $\lambda_i \neq 0$

$$A^{-1} = QD^{-1}Q^{\mathsf{T}}$$

Inverting a diagonal matrix means reciprocating its entries, so the whole inversion costs nothing once the factorization exists. A zero eigenvalue is precisely what makes it fail — the same condition as everywhere else, read off the diagonal.

3 **Square root**

all $\lambda_i \geq 0$ — positive semi-definite

$$A^{1/2} = QD^{1/2}Q^{\mathsf{T}}$$

A symmetric square root exists exactly when no eigenvalue is negative, and it is unique among positive semi-definite matrices. It is a different factor from the triangular one in Cholesky, though both satisfy $A = BB^{\mathsf{T}}$, and it is the one behind whitening a covariance matrix.

4 **Exponential**

always defined

$$e^A = Qe^DQ^{\mathsf{T}}$$

No condition at all, since e^x is defined for every real x. This is what solves the linear system of differential equations $\mathbf{x}' = A\mathbf{x}$ — the solution is $e^{At}\mathbf{x}_0$, and the decomposition makes it computable.

SCALARS, STRAIGHT OFF THE DIAGONAL — 3

5 **Trace**

always

$$\operatorname{tr}(A) = \lambda_1 + \cdots + \lambda_n$$

The trace is similarity invariant, so $\operatorname{tr}(A) = \operatorname{tr}(D)$ and D is diagonal — the sum is immediate.

6 **Determinant**

always

$$\det(A) = \lambda_1\lambda_2\cdots\lambda_n$$

Same argument with the multiplicative property: $\det Q \cdot \det D \cdot \det Q^{\mathsf{T}} = \det D$, since $\det Q = \pm 1$ and the two cancel.

7 **Rank**

always

count of $\lambda_i \neq 0$

Multiplying by an invertible matrix preserves rank, so $\operatorname{rank}(A) = \operatorname{rank}(D)$, which is the number of nonzero diagonal entries. For a symmetric matrix, algebraic and geometric multiplicity always agree — so this count is unambiguous in a way it is not for a general matrix.

The pattern is one rule: $f(A) = Qf(D)Q^{\mathsf{T}}$, and $f(D)$ applies f to each diagonal entry. Powers, inverses, roots and exponentials are all the same statement with a different f, which is what makes a symmetric matrix genuinely easier to work with rather than merely nicer to look at.